MANUFACTURING

Replenishment & MPS — From Excel to Odoo

Reverse-engineering the RAP MPS workbook • Migration strategy to dryer-centric planning

CURRENT STATE
RAP MPS Excel workbook
3 BOM levels × 2 families
Rolling 6-month horizon

CORE LOGIC
Safety stock reorder point
Pack rounding • Waste cascade
MTS vs MTO per SKU

TARGET STATE
Odoo MPS with dryer capacity
Planning families • Lot tracing
Automated replenishment

Kebony — Manufacturing | March 2026

Status: Architecture documentation

1. Current Architecture — RAP MPS Workbook

Three BOM levels × two product families • Each level has a RAD/SP decision sheet + a Calc computation engine

BOM LEVELS & SHEET STRUCTURE

LEVEL	DESCRIPTION	RADIATA	SPRUCE/PINE	CALC
FG	Finished Goods	FG RAD	FG SP	FG Calc
RS	Rough Sawn	RS RAD	—	RS Calc
RM	Raw Material	RM RAD	RM SP	RM Calc

WASTE / YIELD FACTORS

FG → RS

1.10×

10% waste in finishing

RS → RM

1.05×

5% waste in sawing

Overall RM→FG yield: ~86% — 1 / (1.10 × 1.05). Every 100 boards of raw material produce ~86 boards of finished goods.

DEMAND CASCADE FLOW

```
DATA SOURCES                    CALC ENGINES            DECISION SHEETS

Demand Data →─┐
Sales Orders→─└ FG Calc → FG RAD / FG SP
                    ↓                           ↓
                    ↓  ×1.10 waste              ↓
                    ↓                           ↓
               RS Calc → RS RAD
                    ↓                           ↓
                    ↓  ×1.05 waste              ↓
                    ↓                           ↓
               RM Calc → RM RAD / RM SP
                    ↓
               Purchase Orders
```

15 data source sheets feed the workbook: SO, Demand Data, US FC, Prod, Pur, Inv, PP, Safety Stocks, Item Master, Customers, Monthly Split, and more. All pulled from AX2012.

2. SKU Granularity & Data Model

Each FG product explodes into ~4–5 rows (one per foot length) • Maps to Odoo product template → variant structure

RAD SHEET COLUMNS

COL	FIELD	EXAMPLE	MEANING
A	**WW**	1934	Input WW product (raw/semi ref)
B	**RS**	1935	Rough Sawn product reference
C	**Repl. Time**	1	Lead time in months
D	**Part Nr**	2131	FG item number (AX item ID)
E	**FT**	10, 12, 14...	Foot length
F	**M**	30, 36, 42...	Board dimension in mm
H	**Planning Method**	MTS / MTO	Make to Stock vs Order

Same Part Nr, multiple rows. Part Nr 2131 appears as 5 rows: 10 FT, 12 FT, 14 FT, 16 FT, 18 FT. This is the product template → variant mapping in Odoo (FT = variant attribute).

PP SHEET — PACK PARAMETERS

PACK ROUNDING LOGIC

The PP sheet defines **boards per pack** for each WW product. Pack size in linear meters is calculated as:

```
Pack Size = (board_mm / 10) × boards_per_pack(WW)

Example:
  WW = 1934, boards_per_pack = 120
  board_mm = 42
  Pack Size = (42/10) × 120 = 504 LM
```

ODOO MAPPING

```
Excel                  Odoo
Part Nr            →  product.product.default_code
FT (foot length)   →  Product variant attribute
M (dimension mm)   →  Product variant attribute
WW / RS            →  BOM parent ↔ child
Planning Method    →  product.template.route_ids
Boards per Pack    →  kebony.planning.family.pack_boards
```

3. The Replenishment Formula

Classic safety-stock reorder point with pack rounding • Distinct logic for MTS vs MTO items

MAKE TO STOCK (MTS)

```
Step 1: Compute Thresholds
  Safety Stock  = dimension_mm × demand_factor
  Target Stock  = demand_factor × (dimension_mm + 0.75)

Step 2: Check Availability
  On Hand       = SUMIFS(AX inventory snapshot)
  Available     = On Hand − Open SO (current month)

Step 3: Trigger Decision
  IF Available ≤ Safety Stock:
    Produce = ROUNDUP(
      (Target − Available) / Pack Size
    ) × Pack Size
  ELSE:
    Produce = 0
```

Pack rounding is critical. Production quantities must be exact multiples of pack size to match physical stacking and dryer load constraints.

MAKE TO ORDER (MTO)

ZERO SAFETY STOCK

MTO items have **Safety Stock = 0** and **Target Stock = 0**. Production only triggers against confirmed sales demand. No speculative inventory build.

ROLLING FORWARD (M+1 THROUGH M+5)

```
For each forward month:
  Available[M+n] = Available[M+n−1]
                   − Demand[M+n]
                   + Production[M+n]

  IF Available[M+n] ≤ Safety Stock:
    Trigger replenishment (same formula)

Slow Mover Flag:
  IF stock > ½ annual_demand → "SLOW"
  ELSE → "OK"

MOH = inventory / (annual_demand / 12)
```

Existing production orders are subtracted before triggering new production. The formula avoids double-ordering.

4. The Calc Engine — Rolling Projection

7-row repeating block per SKU • 12-month rolling inventory balance • Two-month lookahead in PlndProd

7-ROW BLOCK PER SKU

#	ROW	SOURCE	DESCRIPTION
1	**OpenSales**	So sheet	SUMIFS by product/FT/month
2	**FC**	Demand Data	Historical × growth × monthly split
3	**US FC**	US FC sheet	US forecast × quarterly split %
4	**Prod**	Prod sheet	Existing/confirmed MOs
5	**PlndProd**	Computed	Replenishment trigger (key row)
6	**Inv**	Computed	Rolling inventory balance
7	**MOH**	Computed	Months on Hand

US QUARTERLY SPLIT PERCENTAGES

Q1 (JAN–MAR)	Q2 (APR–JUN)	Q3 (JUL–SEP)	Q4 (OCT–DEC)
5% / 5% / 5%	9% / 9% / 9%	10% / 10% / 10%	9% / 9% / 9%

PLNDPROD FORMULA (TWO-MONTH LOOKAHEAD)

```
PlndProd[month] =
  IF ( Inv[month-1]
       - OpenSales[month]
       - OpenSales[month+1]  ← lookahead
       - FC[month]
       - US_FC[month]
       + Prod[month]
       < Safety_Stock × Repl_Factor
  )
  THEN: MAX(Target - projected_avail, 0)
  ELSE: 0
```

INVENTORY BALANCE (ROLLING)

```
Inv[month] = Inv[month-1]
             - OpenSales[month]
             - FC[month]
             - US_FC[month]
             + Prod[month]        (confirmed)
             + PlndProd[month]    (proposed)
```

GLOBAL PARAMETERS

PARAMETER	VALUE
Growth target	1.2× (Calc) / 1.1× (RAD)
Repl. lead time	1 month (all items)

5. What Works Well — Preserve in Odoo

The RAP MPS workbook embeds proven operational discipline • These patterns must survive the migration

- ✓ **Safety stock reorder point** — Classic trigger: when projected inventory drops below safety stock, produce up to target level
- ✓ **Pack rounding to production multiples** — Quantities always rounded up to full packs, matching physical stacking constraints
- ✓ **MTS vs MTO distinction** — Per-SKU planning method: MTO items get zero safety stock, only produce against demand
- ✓ **Slow mover detection (SLOW flag)** — Flags SKUs where stock exceeds half of annual demand — FIFO discipline mechanism
- ✓ **US forecast overlay with seasonality** — Separate quarterly-split forecast for the US market, additive to base demand
- ✓ **Multi-level BOM explosion with waste** — FG → RS (×1.10) → RM (×1.05) cascade ensures raw material covers yield loss
- ✓ **Rolling 6-month projection** — Forward visibility of inventory balance, not just current month reorder
- ✓ **Existing PO/MO deduction** — Avoids double-ordering by subtracting confirmed production and purchase orders

> **Key insight:** The Excel logic is sound. The problem is not the algorithm — it is the lack of capacity constraints, traceability, and automation. Odoo must replicate this logic AND add what Excel cannot do.

6. What Excel Cannot Do — Odoo Must Add

10 structural limitations of the spreadsheet approach • Each maps to a specific Odoo capability

LIMITATIONS

- ✕ **No dryer capacity constraint** — ignores the primary bottleneck entirely
- ✕ **No family batching** — plans per SKU independently, not by compatible families
- ✕ **No autoclave/dryer load composition** — cannot model process pack → load hierarchy
- ✕ **No WIP tracking** — only sees FG/RS/RM inventory, blind to in-process stock
- ✕ **No lot traceability** — no link from raw material batch to finished goods
- ✕ **No cost integration** — quantities only, no FIFO cost flow
- ✕ **No multi-entity view** — single warehouse, no cross-company planning
- ✕ **Static forecast** — manual refresh, no live SO pipeline
- ✕ **No chemical/mix planning link** — cannot trigger mix batches from tank levels
- ✕ **Approximate pack rounding** — not tied to planning family definitions

ODOO SOLUTIONS

GAP	ODOO CAPABILITY
Dryer capacity	Dryer-centric scheduling on `kebony.planning.family`
Family batching	Planning Family grouping + load composition
Load composition	Process Pack → Autoclave Load → Dryer Load
WIP tracking	Real-time WIP via stock locations + MO states
Lot traceability	Full lot tracking RM → RS → FG
Cost integration	FIFO lot cost flows through BOM explosion
Multi-entity	Multi-company with entity-specific rules
Live forecast	Dynamic from Odoo SO pipeline + CRM
Chemical planning	Mix batch MO triggered by tank reorder point
Pack rounding	Exact calc from `planning.family.pack_boards`

The dryer is the #1 gap. Excel produces plans that are physically infeasible because they ignore the 3-dryer bottleneck (~80 h cycle time per load).

7. Odoo Object Model Mapping

Excel concepts → Odoo models and fields • The translation table for implementation

EXCEL CONCEPT	ODOO MODEL / FIELD	NOTES
Part Nr (item ID)	`product.product.default_code`	Unique product reference, maps 1:1
FT (foot length)	Product variant attribute	Each foot length is a variant of the same template
M (board dimension mm)	Product variant attribute	Dimensional attribute on product template
WW / RS product	BOM parent ↔ child relationship	Multi-level BOM with waste factor in qty ratio
Planning Method (MTS/MTO)	`product.template.route_ids`	MTO route vs MTS route per product
Safety Stock	Reordering rule (min qty)	`stock.warehouse.orderpoint`
Target Stock	Reordering rule (max qty)	Max qty on the same orderpoint
On Hand Stock	`product.product.qty_available`	Real-time from stock quants
Open Sales Orders	`product.product.outgoing_qty`	From confirmed SO lines
Production Orders	`product.product.incoming_qty`	From confirmed MO outputs
Forecast (FC)	Odoo Demand Forecast module	Or custom forecast model
US FC	Sales area-specific forecast overlay	Additive forecast per sales region
Pack Size	`kebony.planning.family.pack_boards`	Boards per pack × board dimensions
Waste Factor	BOM quantity ratio	1.10 RS per 1.00 FG, 1.05 RM per 1.00 RS
SLOW Control	Custom report	Stock vs annual demand ratio analysis
MOH	Custom report	Inventory ÷ monthly consumption

> **Growth Rate** (1.1× / 1.2×) and **US Quarterly Splits** become configurable forecast parameters — not hardcoded in BOM or reordering rules.

8. Migration Phases

Three-phase roadmap from manual Excel planning to fully automated dryer-centric MPS

PHASE 1 — CURRENT

PLANNING FAMILIES IN ODOO

Planning families defined with reference process times

Basic MTS/MTO replenishment rules per product

Safety stock and reorder points configured per SKU

Manual planning using planning family views

EXCEL STILL USED FOR MPS

PHASE 2 — TARGET

FULL AUTOMATED MPS

Automated replenishment scheduler

Safety stock + target stock logic (from RAP MPS)

Pack rounding to family-compatible quantities

Dryer capacity constraint (max loads/week)

Waste factor cascade FG → RS → RM

Rolling 6-month projection view

US forecast overlay integrated

Slow mover report from FIFO lot age

PHASE 3 — FUTURE

OPTIMISATION & INTELLIGENCE

Dryer scheduling optimisation (family sequencing to minimise changeovers)

MES integration (Prediktor actual vs planned cycle times)

AI-assisted demand forecasting replacing static growth factors

REQUIRES PHASE 2 COMPLETE

Phase 2 is the critical milestone. It replaces the RAP MPS workbook entirely. Until Phase 2 is live, Excel remains the system of record for production planning. Parallel running recommended during transition.

9. Decision Points — CEO/CFO Approval Required

Five open questions that must be resolved before Phase 2 implementation can begin

1 Safety Stock Levels Validation

Current safety stock formulas are embedded in Excel. Business must **sign off on the formula** (dimension × demand_factor) before it is codified in Odoo reordering rules. Are these levels correct, or do they need recalibration?

2 Growth Rate Assumptions

The Calc engine uses **1.2×** while the RAD sheets use **1.1×**. Which is the correct growth assumption? This directly impacts safety stock levels and forecast-driven production triggers.

3 US Forecast Ownership

The quarterly split percentages (5/5/5 → 9/9/9 → 10/10/10) are maintained manually. **Who owns this forecast in Odoo?** US sales team? Supply chain? How often is it updated?

4 Waste Factors Confirmation

RS→FG = **10%**, RM→RS = **5%**. Are these measured from production data, or are they assumed? Should they vary by product family or remain global? Waste factors drive raw material purchasing.

5 MTO vs MTS Classification Review

Some products currently classified as MTO may benefit from MTS treatment (and vice versa). A **full product classification review** is needed before Odoo replenishment rules are activated. Reclassification changes safety stock, forecast behaviour, and dryer scheduling priority.

These are not IT decisions. Each one directly impacts inventory investment, service levels, and production capacity utilisation. CEO/CFO sign-off required before Phase 2 development starts.